

Set 38: Energy changes

Set 38: Exercises

Notes

- Identify the following reactions as endothermic or exothermic
 - combustion
 - respiration
 - melting
 - boiling
 - freezing/solidification
 - acid-base neutralisation
 - dissociation of sodium hydroxide in water
- Photosynthesis can be represented by the following reaction
$$6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\ell) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) + 6\text{O}_2(\text{g}) \quad \Delta H = +2803 \text{ kJ}$$
 - How much energy is required to produce 1 mole of glucose?
 - How much energy is required to convert 1 mole of carbon dioxide into glucose?
 - How much energy is required to convert 45.0 g of water into glucose and oxygen?
 - What mass of oxygen is produced when 1540 kJ of energy is consumed?
- The pop test is used to confirm the presence of hydrogen. It is a highly exothermic reaction and can be represented by the following equation:
$$2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\ell) \quad \Delta H = -484 \text{ kJ}$$
 - How much energy is produced when 1 mole of hydrogen is reacted with oxygen?
 - How much energy is produced when 5.00 moles of water is formed?
 - What mass of oxygen is required to produce 255 kJ of energy?
 - Write the equation for the decomposition of water into hydrogen and oxygen and include the enthalpy change in the equation.
- Methanol is used as a reactant in fuel cells. It can be produced using the following reaction:
$$2\text{H}_2(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{CH}_3\text{OH}(\text{g}) + 90 \text{ kJ}$$
 - Rewrite the equation showing the heat of reaction as a ΔH .
 - How much energy will be produced when 100.0 g of hydrogen is consumed?
 - What mass of carbon monoxide is required to produce 25.0 kJ of energy?
 - What mass of methanol will be produced when 555 kJ of energy is produced?
- The formation of photochemical smog occurs when chemicals such as ozone and nitrogen monoxide react in the presence of sunlight.
$$\text{O}_3(\text{g}) + \text{NO}(\text{g}) \rightarrow \text{O}_2(\text{g}) + \text{NO}_2(\text{g})$$
 - Ozone and oxygen are allotropes. What does this mean?
 - This reaction is exothermic. Rewrite the equation to include 'heat' in the reaction.
 - Given that the activation energy is 210 kJ and the heat of reaction is -200 kJ, draw a reaction profile diagram for this reaction.
- Separately list common examples of endothermic and exothermic reactions or processes in and around the home e.g. combustion, hot packs, change of phase. Select the reaction that has the most significance to you. Describe and explain the processes involved. Include an energy profile diagram and equations.
- Resusable heat packs usually contain sodium acetate. Describe how these heat packs work and why they are recharged by heating in boiling water. In your answer include reaction profile diagrams and equations where appropriate.